

Lecture 7: Formal Discrete Diffusion

From CTMC reverse rates to SEDD and GIDD

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Logical dependency

The lecture has one backbone:

$$\text{forward CTMC} \implies \text{true reverse rate} \implies \begin{cases} \text{score entropy for SEDD,} \\ \text{CT-ELBO likelihood view.} \end{cases}$$

Part I: Shared mathematics

- 1 column-vector CTMC notation
- 2 small-time reverse posterior
- 3 continuous-time ELBO

Part II: Model families

- 1 SEDD: score entropy for probability ratios
- 2 GIDD: interpolating marginals and posterior denoising
- 3 training, sampling, and comparison

Outline

- 1 CTMC Backbone
- 2 Continuous-time ELBO
- 3 SEDD
- 4 GIDD
- 5 Comparison

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Finite-state CTMC: column-vector convention

Let one token take values in a finite set

$$\mathcal{V} = \{1, \dots, N\}.$$

We write probability distributions as column vectors. Matrix entries use conditional-probability order:

$R_t(y, x)$ is the transition rate from source x to destination y .

Thus, for $y \neq x$,

$$R_t(y, x) \geq 0, \quad R_t(x, x) = - \sum_{y \neq x} R_t(y, x).$$

Small-time transition

$$P(X_{t+\Delta} = y \mid X_t = x) = \delta_{y,x} + \Delta R_t(y, x) + o(\Delta).$$

Forward equation

$$\dot{p}_t(y) = \sum_{x \in \mathcal{V}} R_t(y, x) p_t(x), \quad \dot{p}_t = R_t p_t.$$

Proof of the forward equation: expansion

Proof.

Use total probability over the previous state at time t :

$$p_{t+\Delta}(y) = \sum_{x \in \mathcal{V}} P(X_{t+\Delta} = y \mid X_t = x) p_t(x).$$

Substitute the small-time transition formula:

$$p_{t+\Delta}(y) = \sum_{x \in \mathcal{V}} (\delta_{y,x} + \Delta R_t(y, x) + o(\Delta)) p_t(x).$$

Since the state space is finite, the $o(\Delta)$ terms can be summed:

$$p_{t+\Delta}(y) = p_t(y) + \Delta \sum_{x \in \mathcal{V}} R_t(y, x) p_t(x) + o(\Delta).$$



Proof of the forward equation: limit

Proof (Cont).

Subtract $p_t(y)$, divide by Δ , and let $\Delta \rightarrow 0$:

$$\frac{p_{t+\Delta}(y) - p_t(y)}{\Delta} = \sum_{x \in \mathcal{V}} R_t(y, x) p_t(x) + \frac{o(\Delta)}{\Delta}.$$

Since $o(\Delta)/\Delta \rightarrow 0$,

$$\dot{p}_t(y) = \lim_{\Delta \rightarrow 0} \frac{p_{t+\Delta}(y) - p_t(y)}{\Delta} = \sum_{x \in \mathcal{V}} R_t(y, x) p_t(x).$$

Collecting all $y \in \mathcal{V}$ gives the column-vector form

$$\dot{p}_t = R_t p_t.$$

When the generator has a fixed shape

Many discrete diffusion models choose

$$R_t = \dot{\sigma}(t)G, \quad \sigma(t) = \int_0^t \dot{\sigma}(s) \, ds.$$

If G is time-independent, then

$$p_t = \exp(\sigma(t)G)p_0.$$

The transition matrix over $[0, t]$ is $\exp(\sigma(t)G)$.

Interpretation

G chooses the allowed token changes. The scalar schedule $\dot{\sigma}(t)$ is an isotropic speed: it scales all rates in G by the same amount.

Convention

The first matrix index is always the destination state and the second index is the source state:

$$R_t(y, x) \text{ is the rate from } x \text{ to } y.$$

This matches the order of the conditional probability $P(X_t = y \mid X_s = x)$ and keeps the forward equation as $\dot{p}_t = R_t p_t$.

Sequence state space

For a length- L sequence,

$$X = (X^{(1)}, \dots, X^{(L)}) \in \mathcal{V}^L.$$

The forward corruption is usually token-wise independent:

$$q_t(Z_t | X) = \prod_{i=1}^L q_t(Z_t^{(i)} | X^{(i)}).$$

Why this still models context

The corruption process factorizes, but the denoising network sees the whole noisy sequence Z_t . It can predict each position using all other positions as context.

Locality of the CTMC

With token-wise generators, a single CTMC jump changes one token. Hence the reverse rate only needs probability ratios between sequences at Hamming distance one.

Reverse posterior: keep the time labels

Fix a clean datum x_0 and write $q_t(z) = q_t(z \mid x_0)$. Let

$$s = t - \Delta, \quad X_s = y, \quad X_t = z.$$

Bayes' rule gives the true one-step reverse posterior:

$$q_{s|t}(y \mid z, x_0) = \frac{P(X_t = z \mid X_s = y)q_s(y \mid x_0)}{q_t(z \mid x_0)}.$$

Why the time labels matter

y lives at the earlier time $s = t - \Delta$, while z lives at the later time t . The term $q_s(y)$ must be Taylor expanded around time t :

$$q_s(y) = q_{t-\Delta}(y) = q_t(y) - \Delta \dot{q}_t(y) + o(\Delta).$$

Reverse posterior: off-diagonal case

Proof.

Assume the earlier state differs from the current state: $y \neq z$. Then

$$P(X_t = z \mid X_{t-\Delta} = y) = \Delta R_t(z, y) + o(\Delta).$$

Using the Taylor expansion of the earlier marginal,

$$q_{t-\Delta}(y) = q_t(y) - \Delta \dot{q}_t(y) + o(\Delta).$$

Therefore

$$q_{t-\Delta|t}(y \mid z, x_0) = \frac{(\Delta R_t(z, y) + o(\Delta))(q_t(y) - \Delta \dot{q}_t(y) + o(\Delta))}{q_t(z)}.$$

The derivative term is multiplied by Δ , so it contributes only $O(\Delta^2)$:

$$q_{t-\Delta|t}(y \mid z, x_0) = \Delta R_t(z, y) \frac{q_t(y)}{q_t(z)} + o(\Delta).$$



Reverse posterior: diagonal case

Proof (Cont).

Now the earlier and current state are the same token value: $y = z$. Then

$$P(X_t = z \mid X_{t-\Delta} = z) = 1 + \Delta R_t(z, z) + o(\Delta),$$

and

$$q_{t-\Delta}(z) = q_t(z) - \Delta \dot{q}_t(z) + o(\Delta).$$

Substitute into Bayes' rule:

$$q_{t-\Delta|t}(z \mid z, x_0) = \frac{(1 + \Delta R_t(z, z) + o(\Delta))(q_t(z) - \Delta \dot{q}_t(z) + o(\Delta))}{q_t(z)}.$$

Expanding to first order,

$$q_{t-\Delta|t}(z \mid z, x_0) = 1 + \Delta \left(R_t(z, z) - \frac{\dot{q}_t(z)}{q_t(z)} \right) + o(\Delta).$$

True reverse rates

The off-diagonal expansion defines the true reverse jump rate from the current state z at time t to the earlier state y at time $t - \Delta$:

$$a_t(y, z) := R_t(z, y) \frac{q_t(y \mid x_0)}{q_t(z \mid x_0)}, \quad y \neq z.$$

The order is destination first: $a_t(y, z)$ is the reverse-time rate from source z to destination y . The diagonal coefficient is

$$a_t(z, z) := R_t(z, z) - \frac{\dot{q}_t(z)}{q_t(z)}.$$

Use the forward equation

$$\dot{q}_t(z) = R_t(z, z)q_t(z) + \sum_{y \neq z} R_t(z, y)q_t(y).$$

Thus

$$a_t(z, z) = - \sum_{y \neq z} R_t(z, y) \frac{q_t(y)}{q_t(z)} = - \sum_{y \neq z} a_t(y, z).$$

A unified isotropic token generator

Let π be a chosen prior distribution on $\mathcal{V}$. Define the matrix

$$\Pi_\pi(y, x) = \pi(y),$$

so every column of Π_π equals the column vector π . The isotropic redraw generator is

$$G_\pi = \Pi_\pi - I.$$

Two important choices

- Uniform corruption: $\pi(b) = 1/N$.
- Absorbing MASK corruption: $\pi = \delta_m$.

In both cases, $R_t = \dot{\sigma}(t)G_\pi$ and

the transition matrix over $[0, t]$ is $\exp(\sigma(t)G_\pi)$.

Why the matrix exponential simplifies

Start from a clean token x_0 , represented by the column one-hot vector e_{x_0} . Decompose it into stationary and transient parts:

$$e_{x_0} = \pi + (e_{x_0} - \pi).$$

For any column vector v ,

$$\Pi_\pi v = \pi(\mathbf{1}^\top v).$$

Hence $\Pi_\pi \pi = \pi$ and

$$G_\pi \pi = (\Pi_\pi - I)\pi = 0.$$

Also, because $\mathbf{1}^\top (e_{x_0} - \pi) = 0$,

$$\Pi_\pi (e_{x_0} - \pi) = 0, \quad G_\pi (e_{x_0} - \pi) = -(e_{x_0} - \pi).$$

Thus $e_{x_0} - \pi$ is a right eigenvector of G_π with eigenvalue -1 .

Using the Taylor series of the matrix exponential

By the Taylor expansion,

$$\exp(\sigma G_{\pi}) = I + \sigma G_{\pi} + \frac{\sigma^2}{2!} G_{\pi}^2 + \cdots.$$

Apply this to the two components:

$$\exp(\sigma G_{\pi})\pi = \pi,$$

because $G_{\pi}^k \pi = 0$ for $k \geq 1$. For the transient component,

$$\exp(\sigma G_{\pi})(e_{x_0} - \pi) = \sum_{k=0}^{\infty} \frac{\sigma^k}{k!} G_{\pi}^k (e_{x_0} - \pi) = \sum_{k=0}^{\infty} \frac{\sigma^k}{k!} (-1)^k (e_{x_0} - \pi).$$

Therefore

$$\exp(\sigma G_{\pi})(e_{x_0} - \pi) = e^{-\sigma} (e_{x_0} - \pi).$$

Closed-form token corruption

Combining the two components gives

$$\exp(\sigma(t)G_\pi)e_{x_0} = \pi + e^{-\sigma(t)}(e_{x_0} - \pi).$$

Equivalently,

$$p_{t|0}(\cdot \mid x_0) = e^{-\sigma(t)}e_{x_0} + (1 - e^{-\sigma(t)})\pi.$$

Interpretation

At noise level $\sigma(t)$, the token keeps its clean identity with weight $e^{-\sigma(t)}$ and is redrawn from the prior π with weight $1 - e^{-\sigma(t)}$.

This single formula covers both the uniform prior and the absorbing MASK prior.

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Forward path: Markov before conditioning

Discretize $0 = t_0 < t_1 < \dots < t_K = 1$ and write $z_i = z_{t_i}$. The forward noising transition itself does not depend on knowing the clean datum:

$$q_{i|i-1}(z_i | z_{i-1}).$$

For the likelihood bound of a particular datum x , however, the proposal is the clean-conditioned forward path:

$$q(z_{0:K} | x) = q_0(z_0 | x) \prod_{i=1}^K q_{i|i-1}(z_i | z_{i-1}).$$

The subtle point

The forward chain is Markov regardless of x . The reverse factors used in the ELBO are posterior factors under $q(z_{0:K} | x)$, so they depend on x .

Reverse posterior only after fixing x

For a fixed clean datum x , define the one-step reverse posterior by Bayes' rule:

$$q_{i-1|i}(z_{i-1} | z_i, x) = \frac{q_{i|i-1}(z_i | z_{i-1})q_{i-1}(z_{i-1} | x)}{q_i(z_i | x)}.$$

Conditioned reverse factorization

$$q(z_{0:K} | x) = q_K(z_K | x) \prod_{i=1}^K q_{i-1|i}(z_{i-1} | z_i, x).$$

Why this is the right object

The ELBO for $\log p_\theta(x)$ uses $q(z_{0:K} | x)$ as the proposal. Therefore the exact reverse factors are x -conditioned denoising posteriors, not unconditional reverse transitions.

Proof of the conditioned reverse factorization

Proof.

Substitute Bayes' rule into the proposed reverse product:

$$\begin{aligned} q_K(z_K | x) \prod_{i=1}^K q_{i-1|i}(z_{i-1} | z_i, x) \\ = q_K(z_K | x) \prod_{i=1}^K \frac{q_{i|i-1}(z_i | z_{i-1}) q_{i-1}(z_{i-1} | x)}{q_i(z_i | x)}. \end{aligned}$$

The marginal factors telescope:

$$q_K \cdot \frac{q_0 q_1 \cdots q_{K-1}}{q_1 q_2 \cdots q_K} = q_0,$$

where every q_i is shorthand for $q_i(z_i | x)$. Thus

$$q_K(z_K | x) \prod_{i=1}^K q_{i-1|i}(z_{i-1} | z_i, x) = q_0(z_0 | x) \prod_{i=1}^K q_{i|i-1}(z_i | z_{i-1}) = q(z_{0:K} | x).$$



What is structural here?

A conditional-path statement

Once x is fixed, the conditional path law $q(z_{0:K} | x)$ has two equivalent factorizations:

$$q_0(z_0 | x) \prod_i q_{i|i-1}(z_i | z_{i-1}) \quad \text{and} \quad q_K(z_K | x) \prod_i q_{i-1|i}(z_{i-1} | z_i, x).$$

Modeling implication

During training, x is known, so the exact posterior $q_{i-1|i}(\cdot | z_i, x)$ can be computed or simplified. During generation, x is unknown, so $p_{i-1|i}^\theta(\cdot | z_i)$ learns to replace this family of x -conditioned posteriors.

Start from a discrete-time ELBO

Use the exact reverse factorization as the variational proposal:

$$q(z_{0:K} | x) = q_K(z_K | x) \prod_{i=1}^K q_{i-1|i}(z_{i-1} | z_i, x).$$

Let the model approximate these reverse posteriors by $p_{i-1|i}^\theta(z_{i-1} | z_i)$.

Discrete ELBO

$$\log p_\theta(x) \geq C_{\log} - \sum_{i=1}^K \mathbb{E}_{z_i \sim q_i(\cdot | x)} D_{\text{KL}} \left(q_{i-1|i}(\cdot | z_i, x) \| p_{i-1|i}^\theta(\cdot | z_i) \right).$$

$$C_{\log} = \mathbb{E}_{q_0(z_0|x)} [\log p_\theta(x | z_0)] - D_{\text{KL}}(q_1(\cdot | x) \| p_1).$$

Discrete ELBO proof: Jensen step

Proof.

Define the model path distribution

$$p_{\theta}(x, z_{0:K}) = p_1(z_K) \prod_{i=1}^K p_{i-1|i}^{\theta}(z_{i-1} | z_i) p_{\theta}(x | z_0).$$

Insert the forward path distribution as a proposal:

$$\log p_{\theta}(x) = \log \sum_{z_{0:K}} q(z_{0:K} | x) \frac{p_{\theta}(x, z_{0:K})}{q(z_{0:K} | x)}.$$

By Jensen's inequality,

$$\log p_{\theta}(x) \geq \mathbb{E}_{q(z_{0:K} | x)} \left[\log \frac{p_{\theta}(x, z_{0:K})}{q(z_{0:K} | x)} \right].$$



Discrete ELBO proof: collecting KL terms

Proof (Cont).

Substitute the two path factorizations:

$$\log \frac{p_{\theta}(x, z_{0:K})}{q(z_{0:K} | x)} = \log p_{\theta}(x | z_0) + \log \frac{p_1(z_K)}{q_1(z_K | x)} + \sum_{i=1}^K \log \frac{p_{i-1|i}^{\theta}(z_{i-1} | z_i)}{q_{i-1|i}(z_{i-1} | z_i, x)}.$$

Taking expectation under $q(z_{0:K} | x)$ gives

$$\begin{aligned} & \mathbb{E}[\log p_{\theta}(x | z_0)] - D_{\text{KL}}(q_1(\cdot | x) \| p_1) \\ & - \sum_{i=1}^K \mathbb{E}_{z_i \sim q_i(\cdot | x)} D_{\text{KL}}(q_{i-1|i}(\cdot | z_i, x) \| p_{i-1|i}^{\theta}(\cdot | z_i)). \end{aligned}$$

This proves the discrete-time ELBO stated on the previous slide.

Why only z_i remains outside the KL

For the i -th reverse term, start from the path expectation:

$$\mathbb{E}_{q(z_{0:K} | x)} \left[\log \frac{p_{i-1|i}^{\theta}(z_{i-1} | z_i)}{q_{i-1|i}(z_{i-1} | z_i, x)} \right].$$

The integrand depends only on (z_{i-1}, z_i) , so marginalize all other variables:

$$\sum_{z_{i-1}, z_i} q(z_{i-1}, z_i | x) \log \frac{p_{i-1|i}^{\theta}(z_{i-1} | z_i)}{q_{i-1|i}(z_{i-1} | z_i, x)}.$$

Now split the pair marginal as

$$q(z_{i-1}, z_i | x) = q_i(z_i | x) q_{i-1|i}(z_{i-1} | z_i, x).$$

Why only z_i remains outside the KL (Cont)

Proof (Cont).

Substitute the pair factorization:

$$\sum_{z_i} q_i(z_i | x) \sum_{z_{i-1}} q_{i-1|i}(z_{i-1} | z_i, x) \log \frac{p_{i-1|i}^\theta(z_{i-1} | z_i)}{q_{i-1|i}(z_{i-1} | z_i, x)}.$$

The inner sum is the negative KL:

$$\begin{aligned} \sum_{z_{i-1}} q_{i-1|i}(z_{i-1} | z_i, x) \log \frac{p_{i-1|i}^\theta(z_{i-1} | z_i)}{q_{i-1|i}(z_{i-1} | z_i, x)} \\ = -D_{\text{KL}}(q_{i-1|i}(\cdot | z_i, x) \| p_{i-1|i}^\theta(\cdot | z_i)). \end{aligned}$$

Therefore only the outer marginal $q_i(z_i | x)$ remains outside the KL.

Small-time model reverse kernel

Let the model reverse CTMC have rate $\widehat{R}_t^\theta(y, z)$ from reverse source z to reverse destination y . Write

$$b_t(y, z) := \widehat{R}_t^\theta(y, z).$$

For $s = t - \Delta$,

$$p_{s|t}^\theta(y | z) = \begin{cases} 1 + \Delta b_t(z, z) + o(\Delta), & y = z, \\ \Delta b_t(y, z) + o(\Delta), & y \neq z, \end{cases}$$

where

$$b_t(z, z) = - \sum_{y \neq z} b_t(y, z).$$

What the ELBO will compare

For each current state z , the true reverse rates $a_t(y, z)$ are compared with the model reverse rates $b_t(y, z)$ over all possible destinations y .

Single-step KL expansion

Use the true and model small-time kernels:

$$q_{s|t}(y | z, x) = \begin{cases} 1 + \Delta a_t(z, z) + o(\Delta), & y = z, \\ \Delta a_t(y, z) + o(\Delta), & y \neq z, \end{cases}$$

$$p_{s|t}^\theta(y | z) = \begin{cases} 1 + \Delta b_t(z, z) + o(\Delta), & y = z, \\ \Delta b_t(y, z) + o(\Delta), & y \neq z. \end{cases}$$

First-order term

$$D_{\text{KL}\Delta}(z) = \Delta \left[a_t(z, z) - b_t(z, z) + \sum_{y \neq z} a_t(y, z) \log \frac{a_t(y, z)}{b_t(y, z)} \right] + o(\Delta).$$

Key point

Each reverse-posterior KL is $O(\Delta)$, so the sum over $1/\Delta$ time steps becomes an integral.

Single-step KL proof: diagonal term

Proof.

Let

$$D_{\text{KL}\Delta}(z) = D_{\text{KL}}(q_{s|t}(\cdot | z, x) \| p_{s|t}^{\theta}(\cdot | z)).$$

The diagonal contribution is

$$(1 + \Delta a_t(z, z) + o(\Delta)) \log \frac{1 + \Delta a_t(z, z) + o(\Delta)}{1 + \Delta b_t(z, z) + o(\Delta)}.$$

Using $\log(1 + \Delta c + o(\Delta)) = \Delta c + o(\Delta)$, this equals

$$\begin{aligned} & (1 + \Delta a_t(z, z) + o(\Delta)) [\Delta(a_t(z, z) - b_t(z, z)) + o(\Delta)] \\ & = \Delta(a_t(z, z) - b_t(z, z)) + o(\Delta). \end{aligned}$$



Single-step KL proof: off-diagonal terms

Proof (Cont).

For $y \neq z$, the contribution is

$$(\Delta a_t(y, z) + o(\Delta)) \log \frac{\Delta a_t(y, z) + o(\Delta)}{\Delta b_t(y, z) + o(\Delta)}.$$

The factor Δ cancels inside the logarithm, so

$$\log \frac{\Delta a_t(y, z) + o(\Delta)}{\Delta b_t(y, z) + o(\Delta)} = \log \frac{a_t(y, z)}{b_t(y, z)} + o(1).$$

Therefore the off-diagonal sum is

$$\Delta \sum_{y \neq z} a_t(y, z) \log \frac{a_t(y, z)}{b_t(y, z)} + o(\Delta).$$

Combining with the diagonal term proves the single-step expansion.

From the KL expansion to the CT-ELBO

Proof (Cont).

Let $t_i - t_{i-1} = \Delta$. The discrete ELBO contains

$$\sum_{i=1}^K \mathbb{E}_{z_i \sim q_{t_i}} D_{\text{KL}\Delta}(z_i).$$

Using the first-order expansion,

$$D_{\text{KL}\Delta}(z_i) = \Delta F_{t_i}(z_i) + o(\Delta),$$

where

$$F_t(z) = a_t(z, z) - b_t(z, z) + \sum_{y \neq z} a_t(y, z) \log \frac{a_t(y, z)}{b_t(y, z)}.$$

Thus

$$\sum_{i=1}^K \mathbb{E}_{q_{t_i}} D_{\text{KL}\Delta} \rightarrow \int_0^1 \mathbb{E}_{z \sim q_t} F_t(z) dt.$$

Substituting this limit into the discrete ELBO gives the CT-ELBO.

CT-ELBO in rate form

Taking the continuous-time limit gives

$$\log p_{\theta}(x) \geq C_{\log} - \int_0^1 \mathbb{E}_{z \sim q_t(\cdot|x)} \left[a_t(z, z) - b_t(z, z) + \sum_{y \neq z} a_t(y, z) \log \frac{a_t(y, z)}{b_t(y, z)} \right] dt.$$

Trainable negative CT-ELBO

After removing terms independent of θ ,

$$\mathcal{L}_{\text{CT}}(\theta) = \int_0^1 \mathbb{E}_{z \sim q_t} \left[\sum_{y \neq z} b_t(y, z) - \sum_{y \neq z} a_t(y, z) \log b_t(y, z) \right] dt + \text{const.}$$

Standard forward/backward-rate form

Substitute

$$a_t(y, z) = R_t(z, y) \frac{q_t(y | x)}{q_t(z | x)}, \quad b_t(y, z) = \hat{R}_t^\theta(y, z).$$

Then

$$\log p_\theta(x) \geq C_{\log} + \int_0^1 \mathbb{E}_{z \sim q_t(\cdot | x)} \left[\hat{R}_t^\theta(z, z) - R_t(z, z) \right. \\ \left. + \sum_{y \neq z} R_t(z, y) \frac{q_t(y | x)}{q_t(z | x)} \log \frac{\hat{R}_t^\theta(y, z) q_t(z | x)}{R_t(z, y) q_t(y | x)} \right] dt.$$

Likelihood-bound view

- If SEDD chooses $\hat{R}_t^\theta(y, z) = R_t(z, y) s_\theta(z, t)_y$, CT-ELBO recovers a diffusion-weighted score entropy objective.
- GIDD chooses $\hat{R}_t^\theta(y, z) = R_t(z, y) q_t(y | x_\theta) / q_t(z | x_\theta)$.

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SEDD: modeling the reverse rate by ratios

SEDD learns the concrete score

$$s_{\theta}(z, t)_y \approx \frac{p_t(y)}{p_t(z)}, \quad y \neq z.$$

The model reverse rate is

$$\widehat{R}_t^{\theta}(y, z) = R_t(z, y)s_{\theta}(z, t)_y, \quad y \neq z.$$

Why this is natural

The true reverse rate is

$$a_t(y, z) = R_t(z, y)\frac{p_t(y)}{p_t(z)}.$$

Thus SEDD separates the known forward edge $R_t(z, y)$ from the unknown probability ratio. This edge is the forward transition from the earlier candidate y to the current token z .

Sequence interpretation

$s_{\theta}(Z_t, t)_{i,y}$ estimates the ratio between the current sequence and the sequence obtained by replacing token i by y .

SEDD forward processes

Take

$$R_t = \dot{\sigma}(t)G_\pi, \quad G_\pi = \Pi_\pi - I.$$

For a clean token x_0 ,

$$p_{t|0}(\cdot \mid x_0) = \exp(\sigma(t)G_\pi)e_{x_0}.$$

Closed form

Using the eigen-decomposition from the CTMC backbone,

$$p_{t|0}(\cdot \mid x_0) = e^{-\sigma(t)}e_{x_0} + (1 - e^{-\sigma(t)})\pi.$$

Examples

- Uniform SEDD: choose π uniform on $\mathcal{V}$.
- Absorbing/MASK SEDD: choose $\pi = \delta_m$.

Score entropy

For a fully supported distribution p , weights $w_{xy} \geq 0$, and positive scores $s_\theta(x)_y > 0$, define

$$L_{SE}(\theta; p, w) = \mathbb{E}_{x \sim p} \sum_{y \neq x} w_{xy} \left[s_\theta(x)_y - \frac{p(y)}{p(x)} \log s_\theta(x)_y + K\left(\frac{p(y)}{p(x)}\right) \right],$$

where

$$K(r) = r(\log r - 1).$$

Nonnegative pointwise gap

Let $r = p(y)/p(x)$. Then

$$s - r \log s + K(r) = r \left(\frac{s}{r} - 1 - \log \frac{s}{r} \right) \geq 0.$$

Consistent minimizer

The unique minimizer is

$$s_\theta^*(x)_y = \frac{p(y)}{p(x)}.$$

From implicit to denoising score entropy

The implicit form contains an unknown ratio:

$$\mathbb{E}_{x \sim p} \sum_{y \neq x} w_{xy} \frac{p(y)}{p(x)} \log s_{\theta}(x)_y.$$

If p is generated by corrupting clean data,

$$p(x) = \sum_{x_0} p(x \mid x_0) p_0(x_0),$$

then the denoising identity gives

$$\mathbb{E}_{x \sim p} \sum_{y \neq x} \frac{p(y)}{p(x)} f(x, y) = \mathbb{E}_{x_0 \sim p_0, x \sim p(\cdot \mid x_0)} \sum_{y \neq x} \frac{p(y \mid x_0)}{p(x \mid x_0)} f(x, y).$$

Benefit

The conditional ratio $p(y \mid x_0)/p(x \mid x_0)$ is computable from the forward corruption kernel, while $s_{\theta}(x)_y$ comes from one network forward pass.

Standalone SEDD loss: denoising score entropy

The SEDD loss can be derived without any likelihood-bound argument. Apply score entropy to the noisy marginal p_t and use the denoising identity:

$$L_{DSE}(\theta; t) = \mathbb{E}_{x_0 \sim p_0, z \sim p_t|0(\cdot|x_0)} \sum_{y \neq z} w_t(z, y) \left[s_\theta(z, t)_y - \frac{p_{t|0}(y | x_0)}{p_{t|0}(z | x_0)} \log s_\theta(z, t)_y \right] + \text{const.}$$

What has been used

Only two ingredients are needed:

- 1 score entropy is minimized at the probability ratio $p_t(y)/p_t(z)$;
- 2 the denoising identity replaces the unknown marginal ratio by the computable conditional ratio.

Diffusion-weighted DSE

For a diffusion process, it is natural to weight only edges that can occur under the forward CTMC. With the column-vector convention, choose

$$w_t(z, y) = R_t(z, y).$$

Then the standalone denoising score-entropy objective becomes

$$L_{DWDSE}(x_0) = \int_0^T \mathbb{E}_{z \sim p_{t|0}} \sum_{y \neq z} R_t(z, y) \left[s_{\theta}(z, t)_y - \frac{p_{t|0}(y | x_0)}{p_{t|0}(z | x_0)} \log s_{\theta}(z, t)_y \right] dt + \text{const.}$$

Important point

This is already a valid SEDD training objective from score-entropy ratio estimation. CT-ELBO is a separate route that explains when the same weighted objective is also the trainable part of a likelihood lower bound.

CT-ELBO connection for DWDSE

Condition on clean x_0 . The true and model reverse rates are

$$a_t(y, z) = R_t(z, y) \frac{p_{t|0}(y | x_0)}{p_{t|0}(z | x_0)}, \quad b_t(y, z) = R_t(z, y) s_\theta(z, t)_y.$$

Substituting into the trainable CT-NELBO gives, up to constants,

$$L_{DWDSE}(x_0) = \int_0^T \mathbb{E}_{z \sim p_{t|0}} \sum_{y \neq z} R_t(z, y) \left[s_\theta(z, t)_y - \frac{p_{t|0}(y | x_0)}{p_{t|0}(z | x_0)} \log s_\theta(z, t)_y \right] dt.$$

How to read the connection

Score entropy gives SEDD a ratio-estimation loss directly. CT-ELBO shows that choosing the diffusion weight $R_t(z, y)$ makes that loss coincide with the rate-level likelihood bound, up to constants.

SEDD sequence training

Sample clean sequences $X_0 \sim p_{\text{data}}$ and time $t \sim U(0, T)$. Then corrupt independently:

$$X_t^{(i)} \sim p_{t|0}^{\text{tok}}(\cdot | X_0^{(i)}).$$

The network outputs positive scores

$$s_\theta(X_t, t) \in \mathbb{R}_+^{L \times N}.$$

Single-sample sequence loss

$$\hat{L}_{DWDSE}^{\text{seq}} = \dot{\sigma}(t) \sum_{i=1}^L \sum_{y \neq X_t^{(i)}} G(X_t^{(i)}, y) \left[s_\theta(X_t, t)_{i,y} - \frac{p_{t|0}^{\text{tok}}(y | X_0^{(i)})}{p_{t|0}^{\text{tok}}(X_t^{(i)} | X_0^{(i)})} \log s_\theta(X_t, t)_{i,y} \right].$$

SEDD sampling: Euler and tau-leaping

For a single token, reverse Euler from t to $s = t - \Delta t$ uses

$$P_{\theta}(X_s = y \mid X_t = z) \approx \begin{cases} \Delta t R_t(z, y) s_{\theta}(z, t)_y, & y \neq z, \\ 1 - \Delta t \sum_{u \neq z} R_t(z, u) s_{\theta}(z, t)_u, & y = z. \end{cases}$$

Tau-leaping for sequences

For each position i ,

$$\pi_i(y) = \begin{cases} \Delta t R_t(z_t^{(i)}, y) s_{\theta}(Z_t, t)_{i,y}, & y \neq z_t^{(i)}, \\ 1 - \Delta t \sum_{u \neq z_t^{(i)}} R_t(z_t^{(i)}, u) s_{\theta}(Z_t, t)_{i,u}, & y = z_t^{(i)}. \end{cases}$$

After clipping and normalization, sample positions independently.

Practical meaning

Euler is rate-based and local in time. Tau-leaping trades exact one-jump CTMC semantics for many parallel token updates.

Discrete Tweedie update

If the forward process is $p_t = e^{\sigma(t)G} p_0$ and we know all ratios

$$r_z(i) = \frac{p_t(i)}{p_t(z)},$$

Bayes' rule and $p_0 = e^{-\sigma(t)G} p_t$ give the one-token identity

$$p_{0|t}(x_0 | z) = \left[e^{-\sigma(t)G} r_z \right]_{x_0} \left[e^{\sigma(t)G} \right]_{z, x_0}.$$

Tweedie tau-leaping

For a step of size $\Delta\sigma_t = \sigma(t) - \sigma(t - \Delta t)$, set $r_i(y) = s_\theta(Z_t, t)_{i,y}$ and $r_i(z_t^{(i)}) = 1$. Then

$$\tilde{\pi}_i(y) = \left[e^{-\Delta\sigma_t G} r_i \right]_y \left[e^{\Delta\sigma_t G} \right]_{z_t^{(i)}, y}.$$

Normalize $\tilde{\pi}_i$ and sample each position.

Why SEDD supports infilling

Let Ω be positions to generate and Ω^c be fixed prompt positions. For two candidate fills z, z' ,

$$\frac{p_t(X_\Omega = z' \mid X_{\Omega^c} = y)}{p_t(X_\Omega = z \mid X_{\Omega^c} = y)} = \frac{p_t(X_\Omega = z', X_{\Omega^c} = y)}{p_t(X_\Omega = z, X_{\Omega^c} = y)}.$$

Consequence

The unconditional score ratio is already the conditional score ratio for changes inside Ω .

Inference rule

Keep prompt positions fixed. Apply Euler or Tweedie tau-leaping only to the generated positions.

Outline

- 1 CTMC Backbone
- 2 Continuous-time ELBO
- 3 SEDD
- 4 GIDD**
- 5 Comparison

GIDD: interpolating marginals

GIDD starts by specifying the marginal forward distribution:

$$q_t(z \mid x) = \alpha_t \delta_{z,x} + \beta_t \pi_t(z), \quad \beta_t = 1 - \alpha_t.$$

Here

$$\alpha_0 = 1, \quad \alpha_1 = 0, \quad \pi_t \in \Delta^{|\mathcal{V}|-1}.$$

Interpretation

At time t , a token is a linear interpolation between the clean one-hot token and a possibly time-varying mixing distribution.

Special case

If $\pi_t = \mathbf{m}$ is the MASK point mass, GIDD reduces to masked diffusion.

Constructing forward transition probabilities

For $0 \leq s \leq t \leq 1$, define

$$\alpha_{t|s} := \frac{\alpha_t}{\alpha_s}, \quad v_{t|s}(b) := \beta_t \pi_t(b) - \frac{\alpha_t}{\alpha_s} \beta_s \pi_s(b).$$

Then set

$$P(X_t = b \mid X_s = a) = \alpha_{t|s} \delta_{b,a} + v_{t|s}(b).$$

Equivalent mixture form

If $\beta_{t|s} = 1 - \alpha_{t|s}$ and $\pi_{t|s} = v_{t|s}/\beta_{t|s}$,

$$P(X_t = b \mid X_s = a) = \alpha_{t|s} \delta_{b,a} + \beta_{t|s} \pi_{t|s}(b).$$

Marginal check

Setting $s = 0$ gives

$$P(X_t = z \mid X_0 = x) = \alpha_t \delta_{z,x} + \beta_t \pi_t(z) = q_t(z \mid x).$$

Proof: transition probabilities are valid

Proof.

First check normalization over destination states. For any source state a ,

$$\sum_b P(X_t = b \mid X_s = a) = \sum_b \left(\frac{\alpha_t}{\alpha_s} \delta_{a,b} + \beta_t \pi_t(b) - \frac{\alpha_t}{\alpha_s} \beta_s \pi_s(b) \right).$$

Since $\sum_b \pi_t(b) = \sum_b \pi_s(b) = 1$,

$$\sum_b P(X_t = b \mid X_s = a) = \frac{\alpha_t}{\alpha_s} + \beta_t - \frac{\alpha_t}{\alpha_s} \beta_s = \frac{\alpha_t}{\alpha_s} (1 - \beta_s) + \beta_t = \alpha_t + \beta_t = 1.$$

Under the admissibility condition $v_{t|s}(b) \geq 0$, all entries are nonnegative. Hence these probabilities define a valid Markov transition law. □

Proof: marginal consistency

Proof (Cont).

At $s = 0$, the boundary conditions give $\alpha_0 = 1$ and $\beta_0 = 0$. Therefore

$$\alpha_{t|0} = \alpha_t, \quad \nu_{t|0}(z) = \beta_t \pi_t(z).$$

Thus

$$P(X_t = z \mid X_0 = x) = \alpha_t \delta_{z,x} + \beta_t \pi_t(z) = q_t(z \mid x).$$

This proves that the constructed CTMC has exactly the prescribed GIDD marginal.

Proof: Chapman–Kolmogorov equation

Proof (Cont).

Let $s \leq t \leq u$ and write

$$P(X_t = c \mid X_s = a) = \alpha_{t|s} \delta_{c,a} + v_{t|s}(c), \quad P(X_u = b \mid X_t = c) = \alpha_{u|t} \delta_{b,c} + v_{u|t}(b).$$

Then

$$\begin{aligned} \sum_c P(X_u = b \mid X_t = c) P(X_t = c \mid X_s = a) &= \alpha_{t|s} \alpha_{u|t} \delta_{a,b} + \alpha_{t|s} v_{u|t}(b) \\ &\quad + \alpha_{u|t} v_{t|s}(b) + v_{u|t}(b) \sum_c v_{t|s}(c). \end{aligned}$$

Because $\sum_c v_{t|s}(c) = 1 - \alpha_{t|s}$, this becomes

$$\alpha_{t|s} \alpha_{u|t} \delta_{a,b} + v_{u|t}(b) + \alpha_{u|t} v_{t|s}(b).$$

Proof: Chapman–Kolmogorov equation (Cont)

Proof (Cont).

The first coefficient telescopes:

$$\alpha_{t|s}\alpha_{u|t} = \frac{\alpha_t}{\alpha_s} \frac{\alpha_u}{\alpha_t} = \frac{\alpha_u}{\alpha_s} = \alpha_{u|s}.$$

For the vector term,

$$v_{u|t} + \alpha_{u|t}v_{t|s} = \left(\beta_u \pi_u - \frac{\alpha_u}{\alpha_t} \beta_t \pi_t \right) + \frac{\alpha_u}{\alpha_t} \left(\beta_t \pi_t - \frac{\alpha_t}{\alpha_s} \beta_s \pi_s \right).$$

The middle terms cancel:

$$v_{u|t} + \alpha_{u|t}v_{t|s} = \beta_u \pi_u - \frac{\alpha_u}{\alpha_s} \beta_s \pi_s = v_{u|s}.$$

Therefore

$$\sum_c P(X_u = b \mid X_t = c)P(X_t = c \mid X_s = a) = P(X_u = b \mid X_s = a).$$

Grid proof and telescoping

For a single grid step, define

$$\dot{\alpha}_t = \frac{\alpha_{t+\Delta}}{\alpha_t}, \quad \dot{v}_t = \beta_{t+\Delta} \pi_{t+\Delta} - \frac{\alpha_{t+\Delta}}{\alpha_t} \beta_t \pi_t.$$

The Chapman–Kolmogorov calculation gives

$$\alpha_{t+\Delta|s} = \dot{\alpha}_t \alpha_{t|s}, \quad v_{t+\Delta|s} = \dot{v}_t + \dot{\alpha}_t v_{t|s}.$$

Iterating,

$$\alpha_{t|s} = \prod_i \frac{\alpha_{t_{i+1}}}{\alpha_{t_i}} = \frac{\alpha_t}{\alpha_s},$$

and

$$v_{t|s} = \sum_i \frac{\alpha_t}{\alpha_{t_{i+1}}} \left(\beta_{t_{i+1}} \pi_{t_{i+1}} - \frac{\alpha_{t_{i+1}}}{\alpha_{t_i}} \beta_{t_i} \pi_{t_i} \right) = \beta_t \pi_t - \frac{\alpha_t}{\alpha_s} \beta_s \pi_s.$$

GIDD forward rate

Let $t \rightarrow t + \Delta$. Taylor expansion of the kernel gives

$$\alpha_{t+\Delta|t} = 1 + \frac{\alpha'_t}{\alpha_t} \Delta + o(\Delta),$$

$$v_{t+\Delta|t}(z) = \left(\beta_t \pi'_t(z) - \frac{\alpha'_t}{\alpha_t} \pi_t(z) \right) \Delta + o(\Delta).$$

Forward rate

$$R_t(z, y) = \frac{\alpha'_t}{\alpha_t} \delta_{z,y} + h_t(z), \quad h_t(z) := \beta_t \pi'_t(z) - \frac{\alpha'_t}{\alpha_t} \pi_t(z).$$

Important structure

For source y and destination z with $y \neq z$, the rate $R_t(z, y) = h_t(z)$ depends on the destination z , not on the source y .

GIDD backward parameterization

GIDD predicts a clean token distribution

$$x_\theta(Z_t, t) \in \Delta^{|\mathcal{V}|-1}.$$

It then defines the model-induced marginal

$$q_t(z \mid x_\theta) = \alpha_t x_\theta(z) + \beta_t \pi_t(z).$$

Posterior-form backward transition

$$p_\theta(z_s = y \mid z_t = z) = P(X_t = z \mid X_s = y) \frac{q_s(y \mid x_\theta)}{q_t(z \mid x_\theta)}.$$

Backward rate

For $y \neq z$,

$$\widehat{R}_t^\theta(y, z) = R_t(z, y) \frac{q_t(y \mid x_\theta)}{q_t(z \mid x_\theta)}.$$

The diagonal is determined by the total rate leaving the current reverse state.

Proof: GIDD backward rate

Proof.

Set $s = t - \Delta$. From the posterior-form backward transition,

$$p_{\theta}(z_s = y \mid z_t = z) = P(X_t = z \mid X_s = y) \frac{q_s(y \mid x_{\theta})}{q_t(z \mid x_{\theta})}.$$

For $y \neq z$,

$$P(X_t = z \mid X_s = y) = \Delta R_t(z, y) + o(\Delta), \quad q_s(y \mid x_{\theta}) = q_t(y \mid x_{\theta}) + o(1).$$

Therefore

$$p_{\theta}(z_s = y \mid z_t = z) = \Delta R_t(z, y) \frac{q_t(y \mid x_{\theta})}{q_t(z \mid x_{\theta})} + o(\Delta).$$

Comparing with

$$p_{\theta}(z_s = y \mid z_t = z) = \Delta \hat{R}_t^{\theta}(y, z) + o(\Delta)$$

gives the off-diagonal backward rate. □

Proof: diagonal backward rate

Proof (Cont).

For $y = z$,

$$p_\theta(z_s = z \mid z_t = z) = 1 + \Delta \hat{R}_t^\theta(z, z) + o(\Delta).$$

Since the backward kernel must sum to one,

$$0 = \sum_y \hat{R}_t^\theta(y, z) = \hat{R}_t^\theta(z, z) + \sum_{y \neq z} R_t(z, y) \frac{q_t(y \mid x_\theta)}{q_t(z \mid x_\theta)}.$$

Hence

$$\hat{R}_t^\theta(z, z) = - \sum_{y \neq z} R_t(z, y) \frac{q_t(y \mid x_\theta)}{q_t(z \mid x_\theta)}.$$

This is the diagonal part used in the CT-NELBO derivation.

GIDD CT-NELBO proof: setup

Proof.

Write

$$q_z = q_t(z \mid x), \quad \tilde{q}_z = q_t(z \mid x_\theta), \quad h_z = h_t(z), \quad w_z = \frac{h_z}{q_z}.$$

For $y \neq z$, the GIDD forward rate and model reverse rate are

$$R_t(z, y) = h_z, \quad \hat{R}_t^\theta(y, z) = h_z \frac{\tilde{q}_y}{\tilde{q}_z}.$$

The CT-ELBO integrand for a fixed current state z is

$$I(z) = \hat{R}_t^\theta(z, z) - R_t(z, z) + \sum_{y \neq z} h_z \frac{q_y}{q_z} \log \frac{\hat{R}_t^\theta(y, z) q_z}{h_z q_y}.$$



Proof (Cont).

From the forward rate,

$$R_t(z, z) = \frac{\alpha'_t}{\alpha_t} + h_z.$$

The model diagonal rate is fixed by the total leaving rate:

$$\hat{R}_t^\theta(z, z) = - \sum_{y \neq z} h_z \frac{\tilde{q}_y}{\tilde{q}_z} = -h_z \frac{1 - \tilde{q}_z}{\tilde{q}_z} = h_z - \frac{h_z}{\tilde{q}_z}.$$

Therefore

$$\hat{R}_t^\theta(z, z) - R_t(z, z) = -\frac{h_z}{\tilde{q}_z} - \frac{\alpha'_t}{\alpha_t}.$$

GIDD CT-NELBO proof: logarithmic term

Proof (Cont).

For $y \neq z$,

$$\log \frac{\hat{R}_t^\theta(y, z) q_z}{h_z q_y} = \log \frac{\tilde{q}_y}{q_y} + \log \frac{q_z}{\tilde{q}_z}.$$

Hence

$$\begin{aligned} \sum_{y \neq z} h_z \frac{q_y}{q_z} \log \frac{\hat{R}_t^\theta(y, z) q_z}{h_z q_y} \\ = w_z \sum_{y \neq z} q_y \left(\log \frac{\tilde{q}_y}{q_y} + \log \frac{q_z}{\tilde{q}_z} \right). \end{aligned}$$

Since

$$\sum_{y \neq z} q_y \log \frac{\tilde{q}_y}{q_y} = -D_{\text{KL}}(q \| \tilde{q}) + q_z \log \frac{\tilde{q}_z}{q_z},$$

the whole logarithmic term is

$$w_z \left[-D_{\text{KL}}(q \| \tilde{q}) + \log \frac{q_z}{\tilde{q}_z} \right].$$

GIDD CT-NELBO proof: combine terms

Proof (Cont).

Combine the diagonal term and the logarithmic term:

$$l(z) = -\frac{\alpha'_t}{\alpha_t} - \frac{h_z}{\tilde{q}_z} + w_z \left[-D_{\text{KL}}(q \parallel \tilde{q}) + \log \frac{q_z}{\tilde{q}_z} \right].$$

Because $h_z = w_z q_z$,

$$\frac{h_z}{\tilde{q}_z} = w_z \frac{q_z}{\tilde{q}_z}.$$

Thus

$$l(z) = -\frac{\alpha'_t}{\alpha_t} - w_z \left[D_{\text{KL}}(q \parallel \tilde{q}) + \frac{q_z}{\tilde{q}_z} - \log \frac{q_z}{\tilde{q}_z} \right].$$

GIDD CT-NELBO proof: weight identity

Proof (Cont).

The weight has expectation

$$\mathbb{E}_{z \sim q}[w_z] = \sum_z q_z \frac{h_z}{q_z} = \sum_z h_z.$$

Using

$$h_z = \beta_t \pi'_t(z) - \frac{\alpha'_t}{\alpha_t} \pi_t(z),$$

we get

$$\sum_z h_z = \beta_t \frac{d}{dt} \sum_z \pi_t(z) - \frac{\alpha'_t}{\alpha_t} \sum_z \pi_t(z) = -\frac{\alpha'_t}{\alpha_t}.$$

Taking $-\mathbb{E}_{z \sim q}[l(z)]$ and substituting this identity gives

$$\mathbb{E}_{z \sim q} w_z \left[D_{\text{KL}}(q \| \tilde{q}) + \frac{q_z}{\tilde{q}_z} - \log \frac{q_z}{\tilde{q}_z} - 1 \right].$$

Proof (Cont).

By definition,

$$D_{\text{IS}}(a||b) = \frac{a}{b} - \log \frac{a}{b} - 1.$$

Therefore the negative CT-ELBO integrand becomes

$$w_t(z, x) [D_{\text{KL}}(q_t(\cdot | x) || q_t(\cdot | x_\theta)) + D_{\text{IS}}(q_t(z | x) || q_t(z | x_\theta))],$$

where

$$w_t(z, x) = \frac{1}{q_t(z | x)} \left(\beta_t \pi'_t(z) - \frac{\alpha'_t}{\alpha_t} \pi_t(z) \right).$$

Integrating over t and $z \sim q_t(\cdot | x)$ gives

$$-\log p_\theta(x) \leq \mathbb{E}_{t, z \sim q_t(\cdot | x)} [w_t(z, x) (D_{\text{KL}}(q_t(\cdot | x) || q_t(\cdot | x_\theta)) + D_{\text{IS}}(q_t(z | x) || q_t(z | x_\theta)))] + C.$$

Mask-only special case

Let $\pi_t = \mathbf{m}$ be the MASK point mass. Then $\pi'_t = 0$ and

$$q_t(m \mid x) = 1 - \alpha_t.$$

The weight becomes

$$w_t(z, x) = -\frac{\alpha'_t}{\alpha_t(1 - \alpha_t)} \delta_{z, m}.$$

The loss reduces to masked reconstruction

When $z = m$,

$$D_{\text{KL}}(q_t(\cdot \mid x) \parallel q_t(\cdot \mid x_\theta)) + D_{\text{IS}}(q_t(m \mid x) \parallel q_t(m \mid x_\theta)) = -\alpha_t \log x_\theta(x).$$

Therefore

$$-\log p_\theta(x) \leq \mathbb{E}_{t, z_t} \left[\frac{\alpha'_t}{1 - \alpha_t} \delta_{z_t, m} \log x_\theta(x \mid Z_t, t) \right] + C.$$

Mask plus uniform schedule

GIDD can keep the final prior as all MASK while injecting uniform noise in the middle:

$$\alpha_t = \frac{1-t}{C_t}, \quad \beta_t \pi_t = \frac{t}{C_t} \mathbf{m} + \frac{c_t}{C_t} \mathbf{u}, \quad C_t = 1 + c_t.$$

Here

$$\mathbf{u} = \frac{1}{N-1}(\mathbf{1} - \mathbf{m}), \quad c_t = B t^{\gamma/2} (1-t)^{\gamma/2}.$$

Choosing the maximum uniform ratio

If the desired maximum uniform mass is p_u , choose

$$B = 2^\gamma \frac{p_u}{1 - p_u}.$$

Then $c_t/(1 + c_t)$ is maximized at $t = 1/2$ with value p_u .

Closed-form weights for mask plus uniform

For the mask plus uniform schedule,

$$c'_t = \frac{\gamma}{2} \frac{1-2t}{t(1-t)} c_t, \quad \frac{\alpha'_t}{\alpha_t} = -\frac{1}{1-t} - \frac{c'_t}{1+c_t}.$$

The numerator of the ELBO weight simplifies to

$$\beta_t \pi'_t - \frac{\alpha'_t}{\alpha_t} \pi_t = \frac{\mathbf{m} + (c_t + (1-t)c'_t)\mathbf{u}}{C_t(1-t)}.$$

Hence

$$w_t(\mathbf{z}, \mathbf{x}) = e_{\mathbf{z}}^{\top} \left[\frac{\mathbf{m} + (c_t + (1-t)c'_t)\mathbf{u}}{(1-t)((1-t)\mathbf{x} + t\mathbf{m} + c_t\mathbf{u})} \right].$$

Training note

The exact ELBO weights can be large near the endpoints. Practical GIDD variants often clip or dynamically reweight them to improve gradient stability.

GIDD sequence training

For each clean sequence $X = (x^{(1)}, \dots, x^{(L)})$:

$$Z_t^{(i)} \sim \text{Cat}(\alpha_t \mathbf{e}_{x^{(i)}} + \beta_t \pi_t).$$

The network predicts

$$x_{\theta}^{(i)}(Z_t, t) \in \Delta^{|\mathcal{V}|-1}.$$

Define

$$q_t^{(i)} = \alpha_t \mathbf{e}_{x^{(i)}} + \beta_t \pi_t, \quad q_{\theta,t}^{(i)} = \alpha_t x_{\theta}^{(i)} + \beta_t \pi_t.$$

Sequence loss

$$\hat{L}_{\text{GIDD}}^{\text{seq}} = \sum_{i=1}^L w_t(z_t^{(i)}, x^{(i)}) \left[D_{\text{KL}}(q_t^{(i)} \| q_{\theta,t}^{(i)}) + D_{\text{IS}}(q_t^{(i)}(z_t^{(i)}) \| q_{\theta,t}^{(i)}(z_t^{(i)})) \right].$$

GIDD sampling

Choose a grid

$$0 \approx t_0 < t_1 < \cdots < t_K \approx 1.$$

Since the practical schedule has $\pi_1 = \mathbf{m}$, initialize from the all-MASK sequence.

Ancestral step

For $s < t$,

$$p_\theta(z_s = y \mid z_t = z) = P(X_t = z \mid X_s = y) \frac{q_s(y \mid x_\theta(Z_t, t))}{q_t(z \mid x_\theta(Z_t, t))}.$$

For the forward transition probability,

$$P(X_t = z \mid X_s = y) = \frac{\alpha_t}{\alpha_s} \delta_{z,y} + \beta_t \pi_t(z) - \frac{\alpha_t}{\alpha_s} \beta_s \pi_s(z).$$

All positions can be updated in parallel after clipping numerical negatives and normalizing.

Self-correction in GIDD

The uniform component means the model sees non-MASK but incorrect tokens during training. This enables post-denoising correction.

One-token correction loop

Given a nearly clean sequence Z , compute

$$x_{\theta}^{(1:L)} = \text{softmax}(f_{\theta}(Z, t)/\tau)$$

at small t . Sample candidates $z'^{(i)} \sim \text{Cat}(x_{\theta}^{(i)})$ and define

$$S = \{i : z'^{(i)} \neq z^{(i)}\}.$$

If S is nonempty, update only

$$j = \arg \max_{i \in S} x_{\theta}^{(i)}(z'^{(i)}), \quad z^{(j)} \leftarrow z'^{(j)}.$$

Reason for one-at-a-time updates

Changing one token reduces the risk of introducing multiple incompatible edits at once.

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SEDD and GIDD side by side

Aspect	SEDD	GIDD
Forward process	General CTMC, often $R_t = \dot{\sigma}(t)G_\pi$ with uniform or MASK prior π .	Specify $q_t = \alpha_t x + \beta_t \pi_t$, then construct a compatible CTMC.
Network output	Concrete score $s_\theta(z, t)_y \approx p_t(y)/p_t(z)$.	Clean token distribution $x_\theta(Z_t, t)$.
Reverse rate	$\hat{R}_t^\theta(y, z) = R_t(z, y)s_\theta(z, t)_y$.	$\hat{R}_t^\theta(y, z) = R_t(z, y)q_t(y x_\theta)/q_t(z x_\theta)$.
Training	Score entropy gives ratio estimation directly; DWDSE uses the reverse-edge forward rate.	Closed-form CT-NELBO with KL plus pointwise IS divergence.
Sampling	Euler, tau-leaping, and Tweedie tau-leaping.	Posterior-form ancestral sampling and optional self-correction.

The conceptual difference

SEDD

SEDD asks the model to learn the ratio that the reverse CTMC needs:

$$\frac{p_t(y)}{p_t(z)}.$$

This directly matches the true reverse rate identity and naturally supports conditional sampling through ratio cancellation.

GIDD

GIDD asks the model to predict the clean token distribution:

$$x_\theta(Z_t, t).$$

The reverse kernel is then obtained by plugging this prediction into the exact posterior formula for the chosen interpolating marginal.

Both use the same reverse-rate identity. SEDD has a standalone score-entropy derivation; CT-ELBO gives its likelihood-bound interpretation. GIDD is derived by plugging its posterior-form rate into CT-ELBO.

Core formulas to remember

- ① True reverse rate:

$$a_t(y, z) = R_t(z, y) \frac{q_t(y | x)}{q_t(z | x)}, \quad y \neq z.$$

- ② Trainable CT-NELBO:

$$\int \mathbb{E}_{z \sim q_t} \left[\sum_{y \neq z} b_t(y, z) - \sum_{y \neq z} a_t(y, z) \log b_t(y, z) \right] dt + \text{const.}$$

- ③ SEDD:

$$b_t(y, z) = R_t(z, y) s_\theta(z, t)_y.$$

- ④ GIDD:

$$q_t(z | x) = \alpha_t \delta_{z, x} + \beta_t \pi_t(z), \quad b_t(y, z) = R_t(z, y) \frac{q_t(y | x_\theta)}{q_t(z | x_\theta)}.$$

Summary

Shared foundation

Discrete diffusion can be treated as a CTMC. The single most important identity is the reverse jump rate:

$$a_t(y, z) = R_t(z, y) \frac{q_t(y | x)}{q_t(z | x)}.$$

Model design choices

- SEDD learns the ratio in this identity and trains it with score entropy; CT-ELBO is an additional likelihood interpretation of the diffusion-weighted version.
- GIDD designs an interpolating marginal and predicts clean distributions, yielding a closed-form weighted CT-NELBO.
- Sequence implementations rely on token-wise corruption but context-aware denoisers.

Practical takeaway

Once the forward CTMC and the reverse-rate parameterization are clear, the training losses and samplers are consequences rather than separate tricks.

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